

4

定積分

関数 $f(x)$ の不定積分の1つを $F(x)$ とするとき

$$\int_a^b f(x) dx = [F(x)]_a^b = F(b) - F(a)$$

上端 b
下端 a

$$\int_a^a f(x) dx = 0$$

$$\int_a^b f(x) dx = -\int_b^a f(x) dx$$

$$\int_a^b f(x) dx + \int_b^c f(x) dx = \int_a^c f(x) dx$$

(例1) 次の定積分を求めよ。

$$\begin{aligned} (1) \quad \int_1^2 3x^2 dx &= [x^3]_1^2 \\ &= 2^3 - 1^3 \\ &= 7 // \end{aligned}$$

$$\begin{aligned} (2) \quad \int_1^3 (x^2 - 3x + 1) dx &= \left[\frac{1}{3}x^3 - \frac{3}{2}x^2 + x \right]_1^3 \\ &= \left(\frac{1}{3} \cdot 3^3 - \frac{3}{2} \cdot 3^2 + 3 \right) - \left(\frac{1}{3} \cdot 1^3 - \frac{3}{2} \cdot 1^2 + 1 \right) \\ &= -\frac{4}{3} // \end{aligned}$$

(別解) $\int_1^3 (x^2 - 3x + 1) dx$

$$\begin{aligned} &= \left[\frac{1}{3}x^3 - \frac{3}{2}x^2 + x \right]_1^3 \\ &= \frac{1}{3}(3^3 - 1^3) - \frac{3}{2}(3^2 - 1^2) + (3 - 1) \\ &= -\frac{4}{3} // \end{aligned}$$

$$\begin{aligned} (3) \quad \int_{-1}^2 (t+1)(t-2) dt & \quad \leftarrow t \text{ で微分} \\ &= \int_{-1}^2 (t^2 - t - 2) dt \quad \leftarrow \text{先に展開} \\ &= \left[\frac{1}{3}t^3 - \frac{1}{2}t^2 - 2t \right]_{-1}^2 \\ &= \frac{1}{3}\{2^3 - (-1)^3\} - \frac{1}{2}\{2^2 - (-1)^2\} - 2\{2 - (-1)\} \\ &= -\frac{9}{2} // \end{aligned}$$

$$\begin{aligned} (4) \quad 2 \int_1^3 (x^3 - x^2) dx - \int_1^3 (2x^3 + x^2) dx & \\ &= \int_1^3 \{2(x^3 - x^2) - (2x^3 + x^2)\} dx \\ &= \int_1^3 -3x^2 dx \\ &= [-x^3]_1^3 \\ &= -(3^3 - 1^3) \\ &= -26 // \end{aligned}$$

(例2) 次の定積分を求めよ。

$$\begin{aligned} (1) \quad \int_0^1 (6x-1) dx + \int_1^2 (6x-1) dx & \\ &= \int_0^2 (6x-1) dx \\ &= [3x^2 - x]_0^2 \\ &= 3 \cdot 2^2 - 2 \\ &= 10 // \end{aligned}$$

$$\begin{aligned} (2) \quad \int_1^2 (x^2 - x) dx - \int_2^3 (x^2 - x) dx & \\ &= \int_1^2 (x^2 - x) dx + \int_3^2 (x^2 - x) dx \\ &= \int_1^3 (x^2 - x) dx \\ &= \left[\frac{1}{3}x^3 - \frac{1}{2}x^2 \right]_1^3 \\ &= \frac{1}{3}(3^3 - 1^3) - \frac{1}{2}(3^2 - 1^2) \\ &= \frac{14}{3} // \end{aligned}$$